

PGM Course Project Proposal — Particle Filters for Inference in Continuous State Space

Mohammed

August 23, 2026

Idea. I study the particle filter as approximate inference on a graphical model in continuous state space, using particle-filter SLAM (FastSLAM and gmapping) as the testbed. Working from existing implementations, I experiment with the sampling choices, the proposal distribution and the resampling scheme, and see how they change the result. SLAM here is written as a dynamic Bayesian network, and Rao-Blackwellization uses the conditional independence of the map features given the trajectory to split it into a sampled trajectory and analytic per-feature updates.

Significance. In a particle filter, the sampling choices, mainly the proposal and the resampling, are what really decide its accuracy, how badly the particles degenerate, and its cost. Testing them tells me which choices actually matter, rather than relying on the defaults. And because these sampling questions arise in any particle filter, what I find should carry beyond this SLAM example.

Deliverable. I build on existing implementations (FastSLAM, gmapping) rather than rewriting them, and the work itself is an experimental study plus a report. I adapt the code so I can swap the proposal (motion-model versus measurement-informed) and the resampling scheme (fixed, or adaptive resampling triggered by the effective sample size, so it resamples only when the particles start to collapse) and compare them on trajectory error (ATE / RMSE), the effective sample size and particle degeneracy, and how many particles or how much runtime a target accuracy needs. The report walks through the DBN framing, the Rao-Blackwellization, and the results.

Learning outcome. By the end I will be able to take a continuous estimation problem, write it as a graphical model, and explain the particle filter as inference on it. Running the experiments myself also gives me a hands-on understanding of how the sampling choices trade off accuracy, degeneracy, and cost, and it carries what the course teaches about inference in the discrete case into the continuous, non-Gaussian one.